

EASTEND CYPRESS, SK, Canada

WMO: 711310

Lat: 49.4378N

Lon: 108.9886W

Elev: 3542

StdP: 12.91

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.4	-15.2	-31.9	1.0	-18.8	-24.1	1.6	-14.3	38.1	10.2	34.2	16.5	9.1	300	0.749

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.2	86.5	60.8	83.1	59.9	79.6	59.0	64.6	77.8	62.7	76.4	61.0	74.7	11.3	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	88.8	69.4	57.9	81.8	67.4	55.8	75.7	65.3	31.5	77.6	30.0	76.1	28.7	74.4	72.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.9	26.1	23.1	-26.6	92.0	4.8	2.8	-30.1	94.0	-32.9	95.7	-35.6	97.2	-39.1	99.3
			-26.4	68.1	5.2	2.5	-30.1	69.9	-33.2	71.4	-36.1	72.8	-39.9	74.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.9	13.7	15.2	25.5	39.2	49.5	57.5	64.8	63.9	54.2	40.3	26.4	15.5		
	DBStd	21.29	14.36	13.75	13.60	9.78	8.25	6.44	6.18	6.98	9.24	9.77	12.04	13.37		
	HDD50	5481	1125	975	759	340	110	11	1	2	58	322	708	1070		
	HDD65	9688	1590	1395	1223	775	481	238	81	106	342	764	1158	1535		
	CDD50	1444	0	0	1	15	95	235	460	432	183	22	1	0		
	CDD65	178	0	0	0	0	2	12	76	71	17	0	0	0		
	CDH74	2302	0	0	0	3	65	171	899	919	242	3	0	0		
	CDH80	673	0	0	0	0	5	33	272	304	59	0	0	0		
	WSAvg	10.9	11.4	10.6	11.2	11.9	12.2	11.3	9.7	9.7	10.2	10.8	10.7	10.9		
	PrecAvg	14.0	0.7	0.5	0.6	1.0	2.0	3.0	2.0	1.5	1.1	0.7	0.6	0.6		
	PrecMax	21.0	2.0	0.9	1.5	2.3	4.6	7.0	4.4	5.2	3.5	3.6	1.1	1.7		
(15) Precipitation	PrecMin	8.3	0.1	0.2	0.0	0.1	0.4	1.3	0.6	0.1	0.0	0.0	0.1	0.0		
	PrecStd	3.0	0.5	0.2	0.4	0.6	1.1	1.3	1.1	1.3	0.9	0.7	0.3	0.4		
	DB	41.3	45.5	60.4	71.6	80.2	84.1	90.1	90.9	85.8	72.6	56.9	43.7			
	MCWB	35.8	37.8	45.9	50.2	55.9	60.7	62.2	60.7	58.2	53.2	44.2	35.8			
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	37.9	39.0	52.0	65.9	75.1	78.6	86.3	87.2	81.2	65.3	50.6	39.0			
	2% MCWB	33.9	33.8	40.9	48.4	54.6	59.4	62.1	60.0	56.5	49.3	41.2	33.8			
	5% DB	35.4	35.5	46.5	60.5	70.1	74.7	83.0	83.3	75.8	59.6	45.5	35.5			
	5% MCWB	32.1	31.8	38.4	45.4	51.9	57.6	61.6	59.1	54.9	46.6	37.8	31.4			
	10% DB	32.3	32.5	41.6	55.3	65.4	71.1	79.2	79.2	70.4	54.8	41.1	32.3			
	10% MCWB	29.5	29.5	35.7	42.7	49.6	56.0	60.6	58.3	52.9	44.3	35.0	29.0			
	0.4% WB	36.9	38.5	46.2	52.4	58.2	65.3	68.4	65.3	61.7	53.9	45.9	37.2			
	0.4% MCDB	40.2	44.0	57.6	68.4	74.0	75.7	81.7	79.2	77.8	68.9	55.6	42.4			
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	34.4	34.4	41.9	49.0	55.5	61.9	65.5	63.1	58.3	50.5	41.8	34.0			
	2% MCDB	37.4	38.3	50.2	63.6	71.4	73.8	79.2	77.8	74.9	62.5	49.6	38.2			
	5% WB	32.1	31.9	38.7	46.4	53.2	59.5	63.7	61.7	56.0	47.6	38.5	31.7			
	5% MCDB	35.2	35.0	45.7	58.8	67.3	71.3	77.1	76.9	72.5	58.0	44.5	35.4			
	10% WB	29.6	29.5	35.9	43.8	51.1	57.3	61.8	60.0	53.8	44.9	35.8	29.1			
	10% MCDB	32.3	32.4	41.4	54.3	62.7	68.4	75.0	75.2	68.7	53.6	40.6	32.1			
	MDBR	15.5	15.8	16.7	20.5	21.3	20.1	23.2	24.1	22.4	20.2	16.7	15.0			
	5% DB	16.6	17.2	23.2	29.1	28.6	25.8	28.2	30.1	29.4	26.7	21.6	17.8			
(36) Mean Daily Temperature Range	MCWBR	13.8	13.5	14.3	15.7	13.4	12.2	11.3	11.2	12.7	14.3	14.1	13.8			
	MCDBR	16.6	17.2	21.8	26.6	25.8	23.5	24.0	25.0	26.6	24.7	20.5	17.4			
	MCWBR	14.1	13.8	14.3	15.1	13.0	12.5	11.4	10.6	12.2	13.8	13.9	13.9			
	MCDBR	14.1	13.8	14.3	15.1	13.0	12.5	11.4	10.6	12.2	13.8	13.9	13.9			
(40) Clear-Sky Solar Irradiance	taub	0.250	0.261	0.280	0.321	0.343	0.335	0.340	0.358	0.313	0.282	0.255	0.243			
	taud	2.421	2.439	2.480	2.387	2.366	2.448	2.442	2.360	2.460	2.535	2.506	2.421			
	Ebn at Noon	263	288	299	294	290	293	290	278	283	276	261	250			
	Edh at Noon	19	24	28	35	37	34	34	35	29	22	18	17			
(44) All-Sky Solar Radiation	RadAvg	423	746	1164	1559	1852	2010	2187	1789	1289	808	466	328			
	RadStd	39	51	69	101	128	126	135	124	107	79	36	26			

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page